Efficient Time Series Forecasting Using TinyML Model

This project explores how we can make time-series classification more efficient using lightweight, TinyML-friendly models across different human activity recognition datasets. The main goal was to understand how preprocessing techniques like filtering and downsampling affect not just accuracy, but also training time and inference speed, especially in scenarios where models need to run on resource-constrained devices.

To ensure a well-rounded evaluation, three widely used datasets - UCI-HAR, PAMAP2, and WISDM - were selected, each offering different levels of complexity and noise. We experimented with simple yet effective models such as Logistic Regression and Decision Trees, along with a compact 1D CNN (TinyCNN), to mimic real-world edge deployment conditions.

Different preprocessing pipelines were carefully designed by combining methods like Butterworth filtering, Savitzky-Golay smoothing, and various downsampling strategies. Through these experiments, it became clear that the right preprocessing steps can significantly improve model performance while also reducing computational cost. TinyCNN consistently delivered the best accuracy, while simpler models proved to be faster and more suitable for low-power environments.

Overall, this work highlights an important balance between accuracy and efficiency, and shows how thoughtful preprocessing and model selection can help build practical and optimized TinyML solutions for real-world time-series applications.

- **Datasets Used:**
    1. UCI-HAR (Human Activity Recognition)
    2. PAMAP2
    3. WISDM

- **Models Used:**
    1. Logistic Regression

```
"LogisticRegression": LogisticRegression(max_iter=200),
```

    2. Decision Tree

```
"DecisionTree": DecisionTreeClassifier(max_depth=15, min_samples_split=10),
```

    3. TinyCNN (Lightweight CNN)

```python
def build_tiny_cnn(input_shape, num_classes):

    model = Sequential([
        Input(shape=input_shape),
        Conv1D(16, 3, activation='relu', input_shape=input_shape),
        MaxPooling1D(2),

        Conv1D(32, 3, activation='relu'),
        MaxPooling1D(2),

        Conv1D(32, 3, activation='relu'),

        Flatten(),
        Dense(32, activation='relu'),
        Dense(num_classes, activation='softmax')
    ])

    model.compile(
        optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )

    return model
```

- **DownSampling Methods Used:**
    1. Naive Downsampling

```python
def downsample_naive(X, factor=2):
    return X[:, ::factor, :]
```

    2. Mean Downsampling

```python
def downsample_mean(X, factor=2):
    N, T, F = X.shape
    T_new = T // factor
    X = X[:, :T_new*factor, :]
    return X.reshape(N, T_new, factor, F).mean(axis=2)
```

- **Filtering Methods Used:**
    1. Butterworth Filter

```python
def butter_filter(X, cutoff=5, fs=50, order=3):
    b, a = butter(order, cutoff / (0.5 * fs), btype='low')
    X_filtered = np.zeros_like(X)

    for i in range(X.shape[0]):
        for j in range(X.shape[2]):
            X_filtered[i, :, j] = filtfilt(b, a, X[i, :, j])

    return X_filtered
```

    2. Savitzky-Golay Filter

```python
def savgol_filtering(X, window=7, poly=2):
    X_filtered = np.zeros_like(X)

    for i in range(X.shape[0]):
        for j in range(X.shape[2]):
            X_filtered[i, :, j] = savgol_filter(X[i, :, j], window, poly)

    return X_filtered
```

- **Pipelines Explored:**
  - Filtering → Downsampling
  - Downsampling → Filtering
  - Filtering Only
  - Downsampling Only
  - No Processing (Raw baseline)

- **Sorted According to:**
  The Results for individual models are sorted according to the scores calculated for them.

  $$\textbf{Score} = \textbf{Accuracy} - 0.1 \cdot \textbf{Inference Time} - 0.000001 \cdot \textbf{Training Time}$$

  Accuracy → primary objective (maximized)
  Inference Time → heavily penalized (important for TinyML edge deployment)
  Training Time → lightly penalized (less critical but still considered)

  This reflects real-world constraints:

  - Edge devices care most about fast inference
  - Training happens offline → less important

- **Results:**

  1. UCI-HAR:

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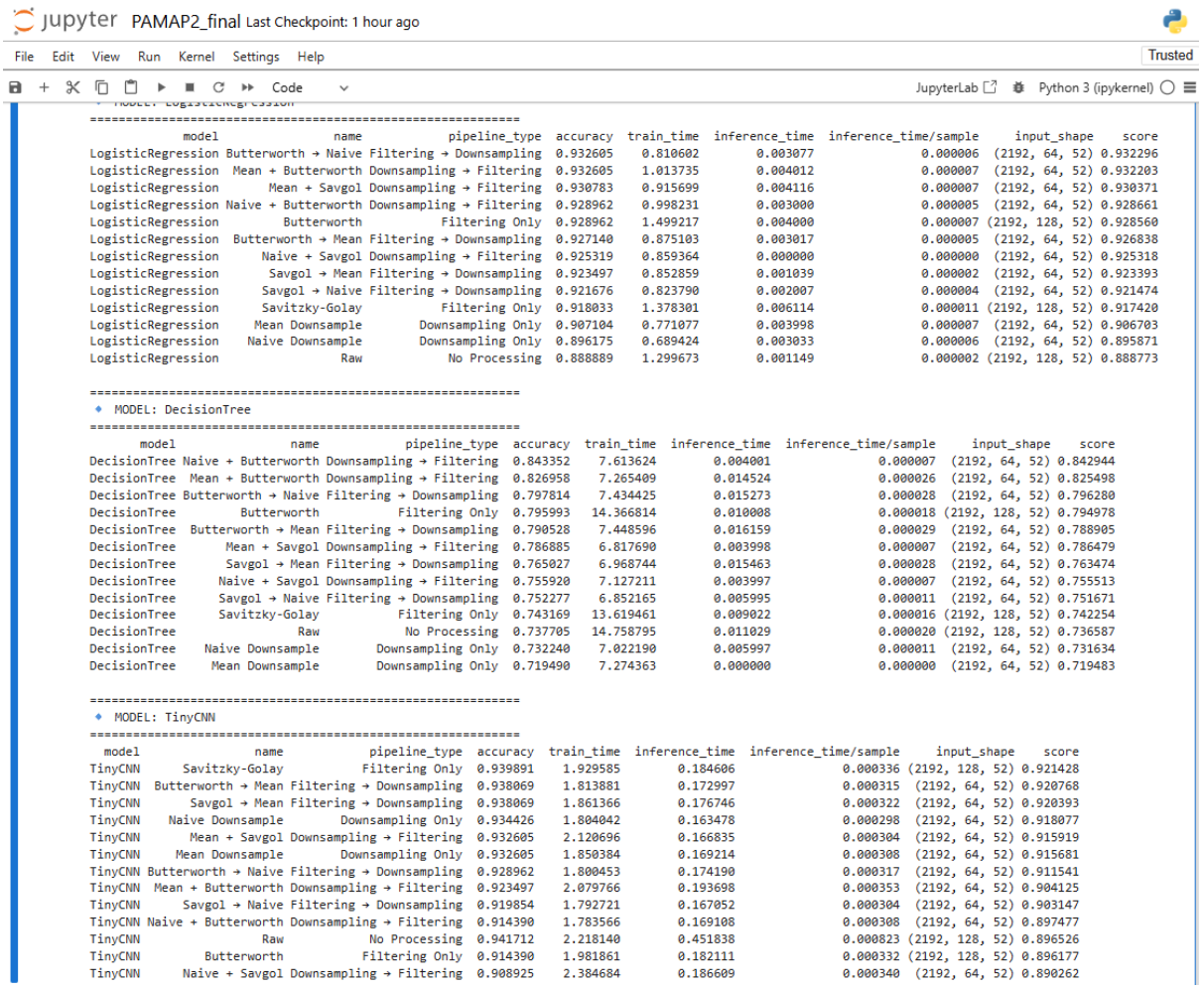
Code JupyterLab Python 3 (ipykernel)

```
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                model                name            pipeline_type  accuracy  train_time  inference_time  inference_time_per_sample    input_shape    score
LogisticRegression                 Raw            No Processing  0.611650    4.680891        0.004026               1.954222e-06 (8239, 128, 9) 0.611243
LogisticRegression      Savitzky-Golay           Filtering Only  0.611650    4.604291        0.005281               2.563694e-06 (8239, 128, 9) 0.611118
LogisticRegression      Savgol → Naive Filtering → Downsampling  0.609223    2.579819        0.002991               1.452154e-06  (8239, 64, 9) 0.608922
LogisticRegression     Mean Downsample        Downsampling Only  0.608738    2.627415        0.002000               9.708034e-07  (8239, 64, 9) 0.608535
LogisticRegression  Butterworth → Mean Filtering → Downsampling  0.608252    2.573138        0.000000               0.000000e+00  (8239, 64, 9) 0.608250
LogisticRegression         Butterworth           Filtering Only  0.607282    4.587848        0.005095               2.473072e-06 (8239, 128, 9) 0.606768
LogisticRegression       Mean + Savgol Downsampling → Filtering  0.605340    2.568325        0.001999               9.704562e-07  (8239, 64, 9) 0.605137
LogisticRegression Butterworth → Naive Filtering → Downsampling  0.605340    2.584699        0.003045               1.478311e-06  (8239, 64, 9) 0.605033
LogisticRegression    Naive Downsample        Downsampling Only  0.604854    2.686532        0.002488               1.207833e-06  (8239, 64, 9) 0.604603
LogisticRegression      Naive + Savgol Downsampling → Filtering  0.603398    2.608900        0.001999               9.705719e-07  (8239, 64, 9) 0.603196
LogisticRegression       Savgol → Mean Filtering → Downsampling  0.601942    2.608191        0.003000               1.456089e-06  (8239, 64, 9) 0.601639
LogisticRegression Naive + Butterworth Downsampling → Filtering  0.597573    2.566074        0.003016               1.463844e-06  (8239, 64, 9) 0.597269
LogisticRegression  Mean + Butterworth Downsampling → Filtering  0.596602    2.641977        0.002999               1.455974e-06  (8239, 64, 9) 0.596299

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🔹 MODEL: DecisionTree
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       model                name            pipeline_type  accuracy  train_time  inference_time  inference_time_per_sample    input_shape    score
DecisionTree  Mean + Butterworth Downsampling → Filtering  0.864078    5.512887        0.002999               0.000001  (8239, 64, 9) 0.863772
DecisionTree Naive + Butterworth Downsampling → Filtering  0.856311    5.628078        0.004009               0.000002  (8239, 64, 9) 0.855904
DecisionTree  Butterworth → Mean Filtering → Downsampling  0.855825    5.289482        0.003000               0.000001  (8239, 64, 9) 0.855520
DecisionTree       Mean + Savgol Downsampling → Filtering  0.849029    5.441875        0.002997               0.000001  (8239, 64, 9) 0.848724
DecisionTree         Butterworth           Filtering Only  0.848058   11.246784        0.006999               0.000003 (8239, 128, 9) 0.847347
DecisionTree      Naive + Savgol Downsampling → Filtering  0.846117    5.425855        0.003220               0.000002  (8239, 64, 9) 0.845789
DecisionTree Butterworth → Naive Filtering → Downsampling  0.839320    5.412191        0.002991               0.000001  (8239, 64, 9) 0.839016
DecisionTree       Savgol → Mean Filtering → Downsampling  0.822330    5.592967        0.004004               0.000002  (8239, 64, 9) 0.821924
DecisionTree     Mean Downsample        Downsampling Only  0.819903    5.557757        0.003008               0.000001  (8239, 64, 9) 0.819597
DecisionTree      Savitzky-Golay           Filtering Only  0.809709   10.992439        0.002163               0.000001 (8239, 128, 9) 0.809481
DecisionTree      Savgol → Naive Filtering → Downsampling  0.808252    5.534255        0.000000               0.000000  (8239, 64, 9) 0.808247
DecisionTree                 Raw            No Processing  0.802427   11.395902        0.005995               0.000003 (8239, 128, 9) 0.801816
DecisionTree    Naive Downsample        Downsampling Only  0.801942    5.445553        0.003000               0.000001  (8239, 64, 9) 0.801636

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🔹 MODEL: TinyCNN
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  model                name            pipeline_type  accuracy  train_time  inference_time  inference_time_per_sample    input_shape    score
TinyCNN      Naive + Savgol Downsampling → Filtering  0.954369    3.545591        0.230448               0.000112  (8239, 64, 9) 0.931321
TinyCNN       Mean + Savgol Downsampling → Filtering  0.954369    3.582596        0.232728               0.000113  (8239, 64, 9) 0.931093
TinyCNN    Naive Downsample        Downsampling Only  0.952913    3.486137        0.231072               0.000112  (8239, 64, 9) 0.929802
TinyCNN     Mean Downsample        Downsampling Only  0.951942    3.521680        0.233193               0.000113  (8239, 64, 9) 0.928619
TinyCNN      Savgol → Naive Filtering → Downsampling  0.950000    3.511932        0.231887               0.000113  (8239, 64, 9) 0.926808
TinyCNN Butterworth → Naive Filtering → Downsampling  0.949515    3.514316        0.230111               0.000112  (8239, 64, 9) 0.926500
TinyCNN         Butterworth           Filtering Only  0.950485    3.819816        0.242782               0.000118 (8239, 128, 9) 0.926203
TinyCNN      Savitzky-Golay           Filtering Only  0.947087    3.838402        0.240408               0.000117 (8239, 128, 9) 0.923043
TinyCNN  Butterworth → Mean Filtering → Downsampling  0.944660    3.472363        0.230958               0.000112  (8239, 64, 9) 0.921561
TinyCNN  Mean + Butterworth Downsampling → Filtering  0.945146    3.899016        0.236595               0.000115  (8239, 64, 9) 0.921482
TinyCNN Naive + Butterworth Downsampling → Filtering  0.938350    3.538521        0.222304               0.000108  (8239, 64, 9) 0.916116
TinyCNN       Savgol → Mean Filtering → Downsampling  0.930583    3.491934        0.234452               0.000114  (8239, 64, 9) 0.907134
TinyCNN                 Raw            No Processing  0.946117    4.159747        0.395513               0.000192 (8239, 128, 9) 0.906561
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2. PAMAP2:



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Code JupyterLab Python 3 (ipykernel)

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	model	name	pipeline_type	accuracy	train_time	inference_time	inference_time/sample	input_shape	score
LogisticRegression	Butterworth → Naive	Filtering → Downsampling	0.932605	0.810602	0.003077	0.000006	(2192, 64, 52)	0.932296
LogisticRegression	Mean + Butterworth	Downsampling → Filtering	0.932605	1.013735	0.004012	0.000007	(2192, 64, 52)	0.932203
LogisticRegression	Mean + Savgol	Downsampling → Filtering	0.930783	0.915699	0.004116	0.000007	(2192, 64, 52)	0.930371
LogisticRegression	Naive + Butterworth	Downsampling → Filtering	0.928962	0.998231	0.003000	0.000005	(2192, 64, 52)	0.928661
LogisticRegression	Butterworth	Filtering Only	0.928962	1.499217	0.004000	0.000007	(2192, 128, 52)	0.928560
LogisticRegression	Butterworth → Mean	Filtering → Downsampling	0.927140	0.875103	0.003017	0.000005	(2192, 64, 52)	0.926838
LogisticRegression	Naive + Savgol	Downsampling → Filtering	0.925319	0.859364	0.000000	0.000000	(2192, 64, 52)	0.925318
LogisticRegression	Savgol → Mean	Filtering → Downsampling	0.923497	0.852859	0.001039	0.000002	(2192, 64, 52)	0.923393
LogisticRegression	Savgol → Naive	Filtering → Downsampling	0.921676	0.823790	0.002007	0.000004	(2192, 64, 52)	0.921474
LogisticRegression	Savitzky-Golay	Filtering Only	0.918033	1.378301	0.006114	0.000011	(2192, 128, 52)	0.917420
LogisticRegression	Mean Downsample	Downsampling Only	0.907104	0.771077	0.003998	0.000007	(2192, 64, 52)	0.906703
LogisticRegression	Naive Downsample	Downsampling Only	0.896175	0.689424	0.003033	0.000006	(2192, 64, 52)	0.895871
LogisticRegression	Raw	No Processing	0.888889	1.299673	0.001149	0.000002	(2192, 128, 52)	0.888773

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🔹 MODEL: DecisionTree
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	model	name	pipeline_type	accuracy	train_time	inference_time	inference_time/sample	input_shape	score
DecisionTree	Naive + Butterworth	Downsampling → Filtering	0.843352	7.613624	0.004001	0.000007	(2192, 64, 52)	0.842944
DecisionTree	Mean + Butterworth	Downsampling → Filtering	0.826958	7.265409	0.014524	0.000026	(2192, 64, 52)	0.825498
DecisionTree	Butterworth → Naive	Filtering → Downsampling	0.797814	7.434425	0.015273	0.000028	(2192, 64, 52)	0.796280
DecisionTree	Butterworth	Filtering Only	0.795993	14.366814	0.010008	0.000018	(2192, 128, 52)	0.794978
DecisionTree	Butterworth → Mean	Filtering → Downsampling	0.790528	7.448596	0.016159	0.000029	(2192, 64, 52)	0.788905
DecisionTree	Mean + Savgol	Downsampling → Filtering	0.786885	6.817690	0.003998	0.000007	(2192, 64, 52)	0.786479
DecisionTree	Savgol → Mean	Filtering → Downsampling	0.765027	6.968744	0.015463	0.000028	(2192, 64, 52)	0.763474
DecisionTree	Naive + Savgol	Downsampling → Filtering	0.755920	7.127211	0.003997	0.000007	(2192, 64, 52)	0.755513
DecisionTree	Savgol → Naive	Filtering → Downsampling	0.752277	6.852165	0.005995	0.000011	(2192, 64, 52)	0.751671
DecisionTree	Savitzky-Golay	Filtering Only	0.743169	13.619461	0.009022	0.000016	(2192, 128, 52)	0.742254
DecisionTree	Raw	No Processing	0.737705	14.758795	0.011029	0.000020	(2192, 128, 52)	0.736587
DecisionTree	Naive Downsample	Downsampling Only	0.732240	7.022190	0.005997	0.000011	(2192, 64, 52)	0.731634
DecisionTree	Mean Downsample	Downsampling Only	0.719490	7.274363	0.000000	0.000000	(2192, 64, 52)	0.719483

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🔹 MODEL: TinyCNN
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	model	name	pipeline_type	accuracy	train_time	inference_time	inference_time/sample	input_shape	score
TinyCNN	Savitzky-Golay	Filtering Only	0.939891	1.929585	0.184606	0.000336	(2192, 128, 52)	0.921428
TinyCNN	Butterworth → Mean	Filtering → Downsampling	0.938069	1.813881	0.172997	0.000315	(2192, 64, 52)	0.920768
TinyCNN	Savgol → Mean	Filtering → Downsampling	0.938069	1.861366	0.176746	0.000322	(2192, 64, 52)	0.920393
TinyCNN	Naive Downsample	Downsampling Only	0.934426	1.804042	0.163478	0.000298	(2192, 64, 52)	0.918077
TinyCNN	Mean + Savgol	Downsampling → Filtering	0.932605	2.120696	0.166835	0.000304	(2192, 64, 52)	0.915919
TinyCNN	Mean Downsample	Downsampling Only	0.932605	1.850384	0.169214	0.000308	(2192, 64, 52)	0.915681
TinyCNN	Butterworth → Naive	Filtering → Downsampling	0.928962	1.800453	0.174190	0.000317	(2192, 64, 52)	0.911541
TinyCNN	Mean + Butterworth	Downsampling → Filtering	0.923497	2.079766	0.193698	0.000353	(2192, 64, 52)	0.904125
TinyCNN	Savgol → Naive	Filtering → Downsampling	0.919854	1.792721	0.167052	0.000304	(2192, 64, 52)	0.903147
TinyCNN	Naive + Butterworth	Downsampling → Filtering	0.914390	1.783566	0.169108	0.000308	(2192, 64, 52)	0.897477
TinyCNN	Raw	No Processing	0.941712	2.218140	0.451838	0.000823	(2192, 128, 52)	0.896526
TinyCNN	Butterworth	Filtering Only	0.914390	1.981861	0.182111	0.000332	(2192, 128, 52)	0.896177
TinyCNN	Naive + Savgol	Downsampling → Filtering	0.908925	2.384684	0.186609	0.000340	(2192, 64, 52)	0.890262

3. WISDM:

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             model                name            pipeline_type  accuracy  train_time  inference_time inference_time/sample     input_shape    score
LogisticRegression      Savgol → Naive Filtering → Downsampling  0.257051    3.098333        0.002002          2.082968e-07  (38434, 64, 3) 0.256847
LogisticRegression      Savitzky-Golay           Filtering Only  0.256530    2.912956        0.001999          2.080239e-07 (38434, 128, 3) 0.256328
LogisticRegression       Savgol → Mean Filtering → Downsampling  0.255906    3.011061        0.000000          0.000000e+00  (38434, 64, 3) 0.255903
LogisticRegression     Mean Downsample        Downsampling Only  0.256530    3.017619        0.008015          8.341549e-07  (38434, 64, 3) 0.255726
LogisticRegression    Naive Downsample        Downsampling Only  0.251327    2.959702        0.001416          1.474081e-07  (38434, 64, 3) 0.251182
LogisticRegression                 Raw           No Processing  0.249766    3.312784        0.002377          2.473261e-07 (38434, 128, 3) 0.249525
LogisticRegression       Mean + Savgol Downsampling → Filtering  0.227391    2.755474        0.001998          2.079494e-07  (38434, 64, 3) 0.227188
LogisticRegression      Naive + Savgol Downsampling → Filtering  0.224997    2.956411        0.000000          0.000000e+00  (38434, 64, 3) 0.224994
LogisticRegression         Butterworth           Filtering Only  0.197003    2.320894        0.010201          1.061631e-06 (38434, 128, 3) 0.195980
LogisticRegression  Butterworth → Mean Filtering → Downsampling  0.195962    2.376245        0.002333          2.428103e-07  (38434, 64, 3) 0.195726
LogisticRegression Butterworth → Naive Filtering → Downsampling  0.195025    2.416992        0.002000          2.080983e-07  (38434, 64, 3) 0.194823
LogisticRegression  Mean + Butterworth Downsampling → Filtering  0.194609    2.361400        0.001999          2.080487e-07  (38434, 64, 3) 0.194407
LogisticRegression Naive + Butterworth Downsampling → Filtering  0.187949    2.238930        0.002000          2.081479e-07  (38434, 64, 3) 0.187747

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🔹 MODEL: DecisionTree
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       model                name            pipeline_type  accuracy  train_time  inference_time inference_time/sample     input_shape    score
DecisionTree     Mean Downsample        Downsampling Only  0.594651    1.073532        0.000378          3.937666e-08  (38434, 64, 3) 0.594612
DecisionTree      Savgol → Naive Filtering → Downsampling  0.590176    1.097852        0.003000          3.122095e-07  (38434, 64, 3) 0.589875
DecisionTree         Butterworth           Filtering Only  0.589447    1.123305        0.003000          3.122095e-07 (38434, 128, 3) 0.589146
DecisionTree    Naive Downsample        Downsampling Only  0.586533    1.108562        0.002000          2.081479e-07  (38434, 64, 3) 0.586332
DecisionTree                 Raw           No Processing  0.586221    1.192212        0.002003          2.084705e-07 (38434, 128, 3) 0.586020
DecisionTree  Butterworth → Mean Filtering → Downsampling  0.581538    1.120319        0.003000          3.122095e-07  (38434, 64, 3) 0.581237
DecisionTree       Mean + Savgol Downsampling → Filtering  0.578520    1.136849        0.003000          3.122095e-07  (38434, 64, 3) 0.578219
DecisionTree      Savitzky-Golay           Filtering Only  0.572276    1.103216        0.002000          2.081231e-07 (38434, 128, 3) 0.572075
DecisionTree       Savgol → Mean Filtering → Downsampling  0.559892    1.108264        0.003000          3.122095e-07  (38434, 64, 3) 0.559591
DecisionTree Butterworth → Naive Filtering → Downsampling  0.551670    1.106791        0.003304          3.437952e-07  (38434, 64, 3) 0.551339
DecisionTree      Naive + Savgol Downsampling → Filtering  0.542720    1.139851        0.003502          3.644636e-07  (38434, 64, 3) 0.542369
DecisionTree  Mean + Butterworth Downsampling → Filtering  0.532626    1.253965        0.003000          3.122095e-07  (38434, 64, 3) 0.532324
DecisionTree Naive + Butterworth Downsampling → Filtering  0.523155    1.132351        0.002172          2.260126e-07  (38434, 64, 3) 0.522937

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🔹 MODEL: TinyCNN
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  model                name            pipeline_type  accuracy  train_time  inference_time inference_time/sample     input_shape    score
TinyCNN    Naive Downsample        Downsampling Only  0.427308   21.234887        0.939858          9.781022e-05  (38434, 64, 3) 0.333301
TinyCNN      Savgol → Naive Filtering → Downsampling  0.423665   19.796777        0.940830          9.791136e-05  (38434, 64, 3) 0.329562
TinyCNN      Savitzky-Golay           Filtering Only  0.417837   21.454415        0.986612          1.026758e-04 (38434, 128, 3) 0.319155
TinyCNN      Naive + Savgol Downsampling → Filtering  0.407326   20.283393        0.913051          9.502041e-05  (38434, 64, 3) 0.316001
TinyCNN       Savgol → Mean Filtering → Downsampling  0.400874   20.668118        0.945390          9.838589e-05  (38434, 64, 3) 0.306315
TinyCNN     Mean Downsample        Downsampling Only  0.374857   20.282817        0.893622          9.299843e-05  (38434, 64, 3) 0.285474
TinyCNN       Mean + Savgol Downsampling → Filtering  0.372151   20.122302        0.902509          9.392332e-05  (38434, 64, 3) 0.281880
TinyCNN                 Raw           No Processing  0.397440   21.445709        1.204105          1.253101e-04 (38434, 128, 3) 0.277008
TinyCNN Butterworth → Naive Filtering → Downsampling  0.360183   19.743803        0.910906          9.479713e-05  (38434, 64, 3) 0.269073
TinyCNN  Butterworth → Mean Filtering → Downsampling  0.361432   19.974839        0.935788          9.738663e-05  (38434, 64, 3) 0.267833
TinyCNN         Butterworth           Filtering Only  0.347487   21.298338        0.988783          1.029017e-04 (38434, 128, 3) 0.248587
TinyCNN Naive + Butterworth Downsampling → Filtering  0.329483   19.657970        0.942295          9.806380e-05  (38434, 64, 3) 0.235234
TinyCNN  Mean + Butterworth Downsampling → Filtering  0.297117   20.680376        0.945387          9.838559e-05  (38434, 64, 3) 0.202558
```